Assignment 1

Total = 15 Marks

Multiple Choice Questions

Question 1:

In the mathematical representation of prediction, $Y = f(x) + \varepsilon$, even if the function $f(\cdot)$ is estimated perfectly, some prediction error may still remain because

(a) predictor variables cannot explain the response
(b) irreducible error cannot be eliminated through model improvement
(c) all models contain estimation bias
(d) additional predictor variables are always required

[1 *Mark*]

Correct answer: (b) irreducible error cannot be eliminated through model improvement
Detailed Solution: For a response variable, the model can be written as

$$Y = f(x) + \varepsilon$$

where $f(x)$ represents the systematic information provided by the predictor variables and ε represents the random error. During model development, the objective is to estimate $f(.)$ so that the reducible error is minimized. However, the irreducible error arises from random variability and therefore cannot be eliminated, even if the function is estimated perfectly.

Question 2:

Identify the statement that correctly describes the role of predictor selection during model development.

(a) Predictor variables should always be selected only on the basis of correlation with the response
(b) Predictor selection is relevant only for statistical models
(c) Physical or conceptual relevance should also be considered while selecting predictor variables
(d) Machine learning models always ignore predictor selection

[2 *Mark*]

Correct answer: (c) Physical or conceptual relevance should also be considered while selecting predictor variables

Detailed Solution: Predictor selection is an important step during model development. Selecting predictors only on the basis of correlation may improve model performance, but physical or conceptual justification is always recommended. Choosing predictors that have

meaningful physical relationships with the response variable generally leads to more reliable and interpretable models.

Question 3:

Interpretability is often considered an advantage of statistical models because they generally

(a) contain fewer parameters that clearly explain the influence of predictor variables
(b) require much larger datasets than machine learning models
(c) eliminate the need for model assumptions
(d) provide perfect predictions for unseen observations

[2 *Mark*]

Correct Answer: (a) contain fewer parameters that clearly explain the influence of predictor variables

Detailed Solution: Statistical models generally contain relatively fewer parameters and predictor variables. As a result, the influence of each predictor on the response variable can often be interpreted directly. Machine learning models may achieve higher prediction accuracy, but they usually involve a large number of parameters, making interpretation more difficult.

Question 4:

A researcher is interested in developing a model for predicting rare extreme events. One important limitation in such studies is that

(a) extreme event data are always categorical
(b) observations corresponding to extreme events are often limited
(c) regression models cannot analyse extreme event data
(d) predictor variables remain constant during extreme events

[1 *Mark*]

Correct answer: (b) observations corresponding to extreme events are often limited

Detailed Solution: Extreme events do not occur frequently. As a result, only a limited number of observations are usually available for model development.

Question 5:

Uncertainty arising due to the natural variability of a process is referred to as

(a) Parameter uncertainty
(b) Operational uncertainty
(c) Aleatory uncertainty
(d) Epistemic uncertainty

[1 *Mark*]

Correct answer: (c) Aleatory uncertainty

Detailed Solution: Aleatory uncertainty originates from the inherent natural variability and randomness present in real-world processes such as rainfall, floods and streamflow.

Question 6:

The primary purpose of uncertainty analysis is to

(a) determine how uncertainties in inputs influence model outputs
(b) eliminate all uncertainties from the model
(c) ensure identical predictions from different models
(d) increase the number of model parameters

[1 *Mark*]

Correct answer: (a) determine how uncertainties in inputs influence model outputs

Detailed Solution: The objective of uncertainty analysis is to quantify the associated uncertainty and assess the confidence that can be placed in model outputs. It also helps determine how uncertainties in input variables and model parameters propagate through the model and influence the final predictions.

Question 7:

A rain gauge records the total rainfall accumulated during every 24-hour period. This is an example of

(a) Sample data representation
(b) Pulse data representation
(c) Categorical data representation
(d) Relative frequency representation

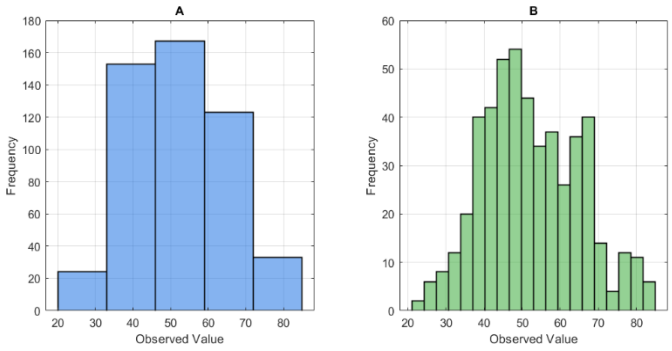
[1 *Mark*]

Correct answer: (b) Pulse data representation

Detailed Solution: In pulse data representation, the recorded value corresponds to the total amount accumulated over a specified time interval. It represents the cumulative effect during that interval rather than the value at a single instant. Precipitation recorded as accumulated rainfall depth over 24 hours is the most common example of pulse data.

Question 8:

The two histograms in the given figure appear noticeably different even though they are constructed from the same observations. The most appropriate explanation is that



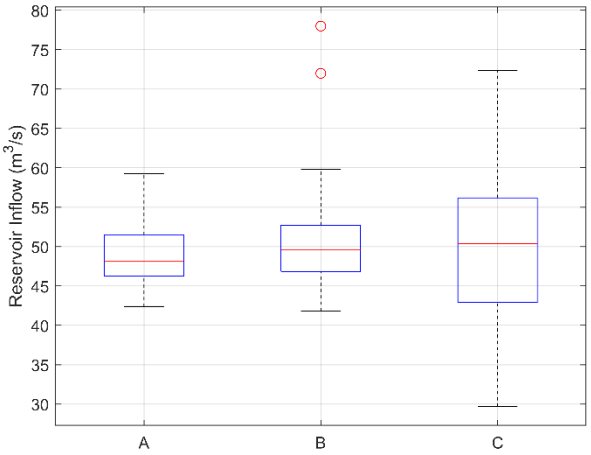
(a) Histograms depend on the choice of bin width and number of bins
(b) The dataset changes whenever the number of bins changes
(c) Histograms cannot be used for continuous variables
(d) Increasing the number of bins changes the mean of the dataset

[2 *Mark*]

Correct answer: (a) Histograms depend on the choice of bin width and number of bins

Detailed Solution: A histogram represents the frequency distribution of numerical data by grouping observations into intervals called bins. The appearance of a histogram depends strongly on the selected number and width of bins. Therefore, different bin selections may produce different visual representations of the same dataset even though the underlying observations remain unchanged.

Question 9:
Based on the boxplots given in the figure, which statement is correct?



(a) Dataset C has the smallest variability
(b) Dataset B contains observations that lie beyond the whiskers
(c) Dataset A has the largest interquartile range
(d) Dataset C must have the highest mean

[2 *Mark*]

Correct answer: (b) Dataset B contains observations that lie beyond the whiskers.

Detailed Solution: A boxplot summarizes the distribution of a dataset by displaying the median, quartiles, interquartile range and possible outliers. Observations that lie beyond the whiskers are identified as outliers. In the given figure, Dataset B contains observations beyond the whiskers, indicating the presence of outliers.

Question 10:

What is the main tradeoff between statistical models and Machine Learning models?

(a) Speed versus computational cost
(b) Interpretability versus accuracy
(c) Data size versus model parameters
(d) Discrete data versus continuous data

[2 *Mark*]

Correct answer: (b) Interpretability versus accuracy

Detailed Solution: The main trade-off between statistical and machine learning models is interpretability versus prediction accuracy. Statistical models generally contain fewer parameters and are easier to interpret, allowing the influence of each predictor variable to be understood. Machine learning models often achieve higher prediction accuracy for large and complex datasets, but their large number of parameters makes interpretation more difficult.